SKILLS TO CAREER
BUILD THE PROOF, LAND THE JOB

+234 802 513 2243
skills to career
stcmentorship1@gmail.com

STC Data Analyst Project – Week 2 Report (FINANCE INDUSTRY)

Name:

Lamidi Habeebullah [Team Head]

Thomeson Emmanuel

Omolara Buhari

INTRODUCTION

Data quality is a critical foundation for any meaningful analysis. Poorly structured or inconsistent data can lead to misleading insights and incorrect decision-making.

This report documents the data cleaning process performed on a consumer complaints dataset. The goal of this process is to identify data quality issues, apply appropriate cleaning techniques, and prepare the dataset for accurate and reliable analysis in subsequent stages of the project.

DATASET OVERVIEW

i. Number of rows before cleaning: 62,516
ii. Number of columns before cleaning: 12
iii. The dataset contains consumer complaints related to financial products and services. Each record represents an individual complaint submitted by a consumer and includes attributes such as:

- Product and sub-product category
- Issue and sub-issue description
- Submission and receipt dates
- Complaint submission channel
- Company public response and private response
- Timeliness of response

The dataset is suitable for analyzing customer dissatisfaction patterns, service performance, and operational efficiency in complaint handling.

TOOL USED

The data cleaning process was carried out using **Microsoft Excel**

Excel was used for:

- Data inspection and filtering
- Handling missing values
- Standardizing categorical variables
- Adding a new column
- Validating data consistency

DATA QUALITY ISSUES IDENTIFIED

A thorough review of the dataset revealed the following issues:

I. Missing Values

Significant missing values were found in several key columns:

- Sub-product: 7 missing entries
- Sub-issue: 10,858 missing entries
- Company public response: 2,175 missing entries
- Timely response: 1,494 missing entries

The high number of missing values in the *Sub-issue* column was particularly notable, as it affects the granularity of analysis.

II. Non-standard column naming

Certain column names were not professionally formatted or analysis-friendly:

- Presence of special characters (e.g., “?”)
- Inconsistent naming conventions

This reduces readability and can cause issues during reporting and visualization.

III. Duplicate Records Check

- A duplicate check was performed, and no duplicate records were found.
- This indicates good data integrity in terms of record uniqueness.

IV. Data Type Validation Issues

Initial inspection suggested that:

- Date fields required confirmation to ensure proper date formatting
- Categorical fields needed consistency for effective grouping

DATA CLEANING PROCESS

The following steps were taken to address the identified issues:

i. Handling Missing Values

To maintain dataset completeness while avoiding data loss:

- Sub-product → Replaced missing values with "Unknown"
- Sub-issue → Replaced missing values with "Not Specified"
- Company public response → Replaced with " Not Specified "
- Timely response? → Replaced with " Not Specified "

Rationale: Instead of deleting rows (which could reduce dataset size and bias analysis), placeholder values were used to retain all observations while clearly identifying missing information.

ii. Column Renaming and Structuring

Columns were renamed to improve clarity and professionalism:

- Timely Response? – Timely Response

Rationale: Clear and standardized column names improve readability and make the dataset easier to use in dashboards and reports.

iii. Data Type Validation

- Verified that date columns are in proper date format
- Confirmed categorical fields are properly recognized as text

Rationale: Correct data types are essential for time-series analysis and accurate computations.

iv. Duplicate Validation

- Conducted a full duplicate check
- Confirmed no duplicate rows exist

CHANGES TO DATASET STRUCTURE

- Standardized column naming convention
- Removed special characters from column headers
- Improved formatting for readability
- Retained all columns as they are relevant to analysis

No columns were dropped to preserve the richness of the dataset.

DATASET READINESS FOR ANALYSIS

Status: YES

The dataset is clean and ready for analysis. After cleaning, the dataset is:

- Free from missing values (handled appropriately)
- Structurally consistent
- Logically validated
- Ready for exploratory data analysis (EDA) and visualization

ADDITIONAL NOTES

Assumptions Made

- Missing categorical values were replaced with descriptive placeholders rather than removing records
- Placeholder values were assumed not to distort overall trends significantly
- The full state name column was added with the assumption that the abbreviation under the state column belongs to the United States of America.

Challenges Encountered

- High volume of missing values in the Sub-issue column required careful handling
- Ensuring consistency across multiple categorical fields demanded detailed inspection

Observations and Future Analysis Opportunities

The dataset provides strong potential for advanced analysis, including:

- Identification of the most common complaint categories
- Trend analysis of complaints over time
- Evaluation of company response performance
- Analysis of response timeliness across different channels
- Customer experience insights based on complaint patterns

.